A **group** is an algebraic structure that occurs in many areas of mathematics. A group is a set G together with a special type of function $*$ from $G \times G$ to G. The function $*$ is called a **binary operation** on G, and it is customary to use the notation $a * b$ instead of $*(a, b)$ to represent a value of $*$. We now give the formal definition of a group. A **group** is a set G together with a binary operation $*$ on G, which satisfies the following three properties:

(i) for all elements a, b, and c in G, $a * (b * c) = (a * b) * c$,

(ii) there exists an element e in G such that $e * a = a * e = a$ for all a in G, and

(iii) for every element a in G there exists an element b in G such that $a * b = b * a = e$. (The element b is called the **inverse** of a in G and is denoted by a^{-1}.)

Let $\text{Isom}_+(\mathbf{E})$ denote the set of all complex functions of the form $f(z) = az + b$ where $|a| = 1$. In the remaining part of this project you are asked to demonstrate that $\text{Isom}_+(\mathbf{E})$ is a group with composition of functions as the binary operation. This group is called the **group of orientation-preserving isometries of the Euclidean plane**.

(**b**) Prove that composition of functions is a binary operation on $\text{Isom}_+(\mathbf{E})$. That is, prove that if f and g are functions in $\text{Isom}_+(\mathbf{E})$, then the function $f \circ g$ defined by $f \circ g(z) = f(g(z))$ is an element in $\text{Isom}_+(\mathbf{E})$.

(**c**) Prove that the set $\text{Isom}_+(\mathbf{E})$ with composition satisfies property (i) of a group.

(**d**) Prove that the set $\text{Isom}_+(\mathbf{E})$ with composition satisfies property (ii) of a group. That is, show that there exists a function e in $\text{Isom}_+(\mathbf{E})$ such that $e \circ f = f \circ e = f$ for all functions f in $\text{Isom}_+(\mathbf{E})$.

(**e**) Prove that the set $\text{Isom}_+(\mathbf{E})$ with composition satisfies property (iii) of a group.

2.4 Special Power Functions

INTRODUCTION A **complex polynomial function** is a function of the form $p(z) = a_n z^n + a_{n-1} z^{n-1} + \ldots + a_1 z + a_0$ where n is a positive integer and a_n, a_{n-1}, $\ldots$, a_1, a_0 are complex constants. In general, a complex polynomial mapping can be quite complicated, but in many special cases the action of the mapping is easily understood. For instance, the complex linear functions studied in Section 2.3 are complex polynomials of degree $n = 1$.

In this section we study complex polynomials of the form $f(z) = z^n$, $n \geq 2$. Unlike the linear mappings studied in the previous section, the mappings $w = z^n$, $n \geq 2$, do not preserve the basic shape of every figure in the complex plane. Associated to the function z^n, $n \geq 2$, we also have the *principal nth root function* $z^{1/n}$. The principal nth root functions are inverse functions of the functions z^n defined on a sufficiently restricted domain. Consequently, complex mappings associated to z^n and $z^{1/n}$ are closely related.

Power Functions

Recall that a real function of the form $f(x) = x^a$, where a is a real constant, is called a power function. We form a complex power function by allowing the input or the exponent a to be a complex number. In other words, a **complex power function** is a function of the form $f(z) = z^\alpha$ where α is a complex constant. If α is an integer, then the power function z^α can be evaluated using the algebraic operations on complex numbers from Chapter 1. For example, $z^2 = z \cdot z$ and $z^{-3} = \dfrac{1}{z \cdot z \cdot z}$. We can also use the formulas for taking roots of complex numbers from Section 1.4 to define power functions with fractional exponents of the form $1/n$. For instance, we *can* define $z^{1/4}$ to be the function that gives the principal fourth root of z. In this section we will restrict our attention to special complex power functions of the form z^n and $z^{1/n}$ where $n \geq 2$ and n is an integer. More complicated

complex power functions such as $z^{\sqrt{2}-i}$ will be discussed in Section 4.2 following the introduction of the complex logarithmic function.

2.4.1 The Power Function z^n

In this subsection we consider complex power functions of the form z^n, $n \geq 2$. It is natural to begin our investigation with the simplest of these functions, the **complex squaring function** z^2.

The Function z^2 Values of the complex power function $f(z) = z^2$ are easily found using complex multiplication. For example, at $z = 2 - i$, we have $f(2-i) = (2-i)^2 = (2-i)\cdot(2-i) = 3-4i$. Understanding the complex mapping $w = z^2$, however, requires a little more work. We begin by expressing this mapping in exponential notation by replacing the symbol z with $re^{i\theta}$:

$$w = z^2 = \left(re^{i\theta}\right)^2 = r^2 e^{i2\theta}. \tag{1}$$

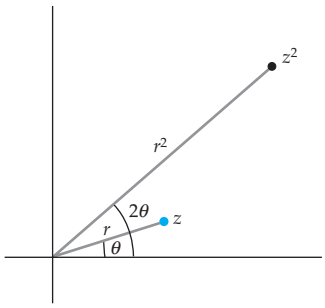
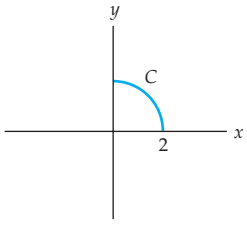


FIGURE 2.4.1 The mapping $w = z^2$

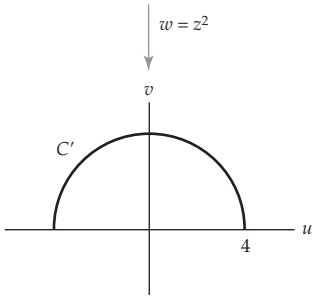
From (1) we see that the modulus r^2 of the point w is the square of the modulus r of the point z, and that the argument 2θ of w is twice the argument θ of z. If we plot both z and w in the same copy of the complex plane, then w is obtained by magnifying z by a factor of r and then by rotating the result through the angle θ about the origin. In FIGURE 2.4.1 we depict the relationship between z and $w = z^2$ when $r > 1$ and $\theta > 0$. If $0 < r < 1$, then z is contracted by a factor of r, and if $\theta < 0$, then the rotation is clockwise.

It is important to note that the magnification or contraction factor and the rotation angle associated to $w = f(z) = z^2$ depend on where the point z is located in the complex plane. For example, since $f(2) = 4$ and $f(i/2) = -\frac{1}{4}$, the point $z = 2$ is magnified by 2 but not rotated, whereas the point $z = i/2$ is contracted by $\frac{1}{2}$ and rotated through $\pi/2$. In general, the squaring function z^2 does not magnify the modulus of points on the unit circle $|z| = 1$ and it does not rotate points on the positive real axis.

The description of the mapping $w = z^2$ in terms of a magnification and rotation can be used to visualize the image of some special sets. The following example illustrates this.



(a) The circular arc of $|z| = 2$



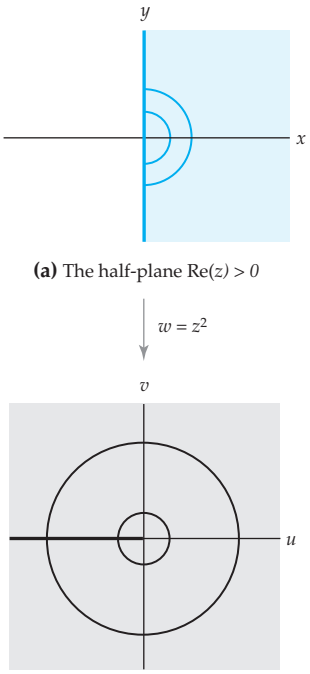
(b) The image of C

FIGURE 2.4.2 The mapping $w = z^2$

EXAMPLE 1 Image of a Circular Arc under $w = z^2$

Find the image of the circular arc defined by $|z| = 2$, $0 \leq \arg(z) \leq \pi/2$, under the mapping $w = z^2$.

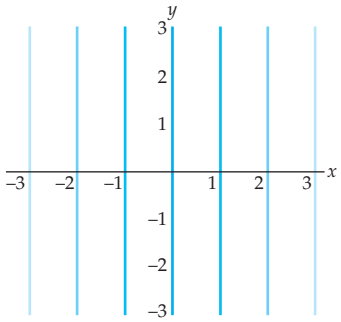
Solution Let C be the circular arc defined by $|z| = 2$, $0 \leq \arg(z) \leq \pi/2$, shown in color in FIGURE 2.4.2(a), and let C' denote the image of C under $w = z^2$. Since each point in C has modulus 2 and since the mapping $w = z^2$ squares the modulus of a point, it follows that each point in C' has modulus $2^2 = 4$. This implies that the image C' must be contained in the circle $|w| = 4$ centered at the origin with radius 4. Since the arguments of the points in C take on every value in the interval $[0, \pi/2]$ and since the mapping $w = z^2$ doubles the argument of a point, it follows that the points in C' have arguments that take on every value in the interval $[2\cdot 0, 2\cdot(\pi/2)] = [0, \pi]$. That is, the set C' is the semicircle defined by $|w| = 4$, $0 \leq \arg(w) \leq \pi$. In conclusion, we have shown that $w = z^2$ maps the circular arc C shown in color in Figure 2.4.2(a) onto the semicircle C' shown in black in Figure 2.4.2(b).



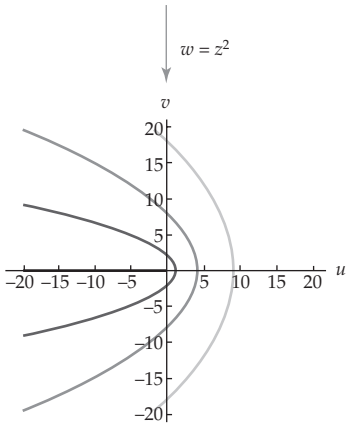
(a) The half-plane Re$(z) > 0$

(b) Image of the half-plane in (a)

FIGURE 2.4.3 The mapping $w = z^2$



(a) Vertical lines in the z-plane



(b) The images of the lines in (a)

FIGURE 2.4.4 The mapping $w = z^2$

An alternative way to find the image in Example 1 would be to use a parametrization. From (10) of Section 2.2, the circular arc C can be parametrized by $z(t) = 2e^{it}$, $0 \le t \le \pi/2$, and by (11) of Section 2.2 its image C' is given by $w(t) = f(z(t)) = 4e^{i2t}$, $0 \le t \le \pi/2$. By replacing the parameter t in $w(t)$ with the new parameter $s = 2t$, we obtain $W(s) = 4e^{is}$, $0 \le s \le \pi$, which is a parametrization of the semicircle $|w| = 4$, $0 \le \arg(w) \le \pi$.

In a similar manner, we find that the squaring function maps a semicircle $|z| = r$, $-\pi/2 \le \arg(z) \le \pi/2$, onto a circle $|w| = r^2$. Since the right half-plane $\text{Re}\,(z) \ge 0$ consists of the collection of semicircles $|z| = r$, $-\pi/2 \le \arg(z) \le \pi/2$, where r takes on every value in the interval $[0, \infty)$, we have that the image of this half-plane consists of the collection of circles $|w| = r^2$ where r takes on any value in $[0, \infty)$. This implies that $w = z^2$ maps the right half-plane $\text{Re}\,(z) \ge 0$ onto the entire complex plane. We illustrate this property in **FIGURE 2.4.3**. Observe that the images of the two semicircles centered at 0 shown in color in Figure 2.4.3(a) are the two circles shown in black in Figure 2.4.3(b). Since $w = z^2$ squares the modulus of a point, the semicircle with smaller radius in Figure 2.4.3(a) is mapped onto the circle with smaller radius in Figure 2.4.3(b), while the semicircle with larger radius in Figure 2.4.3(a) is mapped onto the circle with larger radius in Figure 2.4.3(b). We also see in Figure 2.4.3 that the ray emanating from the origin and containing the point i and the ray emanating from the origin and containing the point $-i$ are both mapped onto the nonpositive real axis. Thus, the imaginary axis shown in color in Figure 2.4.3(a) is mapped onto the set consisting of the point $w = 0$ together with the negative u-axis shown in black in Figure 2.4.3(b).

In order to gain a deeper understanding of the mapping $w = z^2$ we next consider the images of vertical and horizontal lines in the complex plane.

EXAMPLE 2 Image of a Vertical Line under $w = z^2$

Find the image of the vertical line $x = k$ under the mapping $w = z^2$.

Solution In this example it is convenient to work with real and imaginary parts of $w = z^2$ which, from (1) in Section 2.1, are $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$, respectively. Since the vertical line $x = k$ consists of the points $z = k + iy$, $-\infty < y < \infty$, it follows that the image of this line consists of all points $w = u + iv$ where

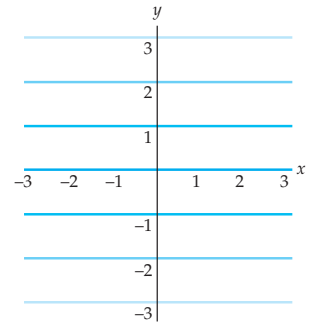
$$u = k^2 - y^2, \quad v = 2ky, \quad -\infty < y < \infty. \tag{2}$$

If $k \neq 0$, then we can eliminate the variable y from (2) by solving the second equation for $y = v/2k$ and then substituting this expression into the remaining equation and inequality. After simplification, this yields:

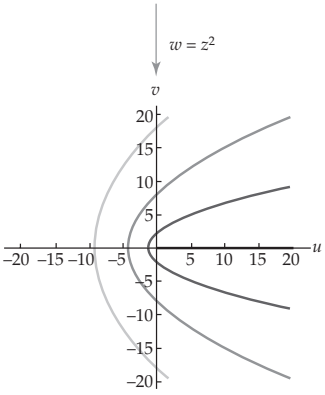
$$u = k^2 - \frac{v^2}{4k^2}, \quad -\infty < v < \infty. \tag{3}$$

Thus, the image of the line $x = k$ (with $k \neq 0$) under $w = z^2$ is the set of points in the w-plane satisfying (3). This image is a parabola that opens in the direction of the negative u-axis, has its vertex at $\left(k^2, 0\right)$, and has v-intercepts at $\left(0, \pm 2k^2\right)$. Notice that the image given by (3) is unchanged if k is replaced by $-k$. This implies that if $k \neq 0$, then the pair of vertical lines $x = k$ and $x = -k$ are both mapped onto the parabola $u = k^2 - v^2/\left(4k^2\right)$ by $w = z^2$.

The action of the mapping $w = z^2$ on vertical lines is depicted in **FIGURE 2.4.4**. The vertical lines $x = k$, $k \neq 0$, shown in color in Figure 2.4.4(a) are mapped onto the parabolas shown in black in Figure 2.4.4(b). In particular, from (3) we have that the lines $x = 3$ and $x = -3$ shown in color in Figure 2.4.4(a) are mapped onto the parabola with vertex at (9, 0) shown in black in Figure 2.4.4(b). In a similar manner,

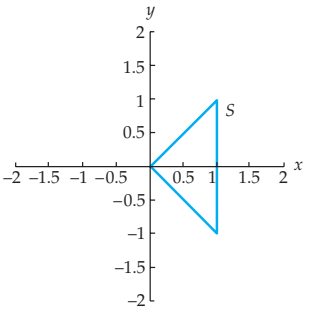


(a) Horizontal lines in the z-plane

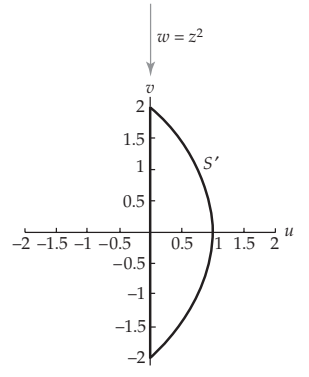


(b) The images of the lines in (a)

FIGURE 2.4.5 The mapping $w = z^2$



(a) A triangle in the z-plane



(b) The image of the triangle in (a)

FIGURE 2.4.6 The mapping $w = z^2$

the lines $x = \pm 2$ are mapped onto the parabola with vertex at $(4, 0)$, and the lines $x = \pm 1$ are mapped onto the parabola with vertex at $(1, 0)$. In the case when $k = 0$, it follows from (2) that the image of the line $x = 0$ (which is the imaginary axis) is given by:

$$u = -y^2, \quad v = 0, \quad -\infty < y < \infty.$$

Therefore, we also have that the imaginary axis is mapped onto the negative real axis by $w = z^2$. See Figure 2.4.4. ☰

With minor modifications, the method of Example 2 can be used to show that a horizontal line $y = k$, $k \neq 0$, is mapped onto the parabola

$$u = \frac{v^2}{4k^2} - k^2 \tag{4}$$

by $w = z^2$. Again we see that the image in (4) is unchanged if k is replaced by $-k$, and so the pair of horizontal lines $y = k$ and $y = -k$, $k \neq 0$, are both mapped by $w = z^2$ onto the parabola given by (4). If $k = 0$, then the horizontal line $y = 0$ (which is the real axis) is mapped onto the positive real axis. Therefore, the horizontal lines $y = k$, $k \neq 0$, shown in color in **FIGURE 2.4.5(a)** are mapped by $w = z^2$ onto the parabolas shown in black in Figure 2.4.5(b). Specifically, the lines $y = \pm 3$ are mapped onto the parabola with vertex at $(-9, 0)$, the lines $y = \pm 2$ are mapped onto the parabola with vertex at $(-4, 0)$, and the lines $y = \pm 1$ are mapped onto the parabola with vertex at $(-1, 0)$.

EXAMPLE 3 Image of a Triangle under $w = z^2$

Find the image of the triangle with vertices 0, $1 + i$, and $1 - i$ under the mapping $w = z^2$.

Solution Let S denote the triangle with vertices at 0, $1+i$, and $1-i$, and let S' denote its image under $w = z^2$. Each of the three sides of S will be treated separately. The side of S containing the vertices 0 and $1+i$ lies on a ray emanating from the origin and making an angle of $\pi/4$ radians with the positive x-axis. By our previous discussion, the image of this segment must lie on a ray making an angle of $2(\pi/4) = \pi/2$ radians with the positive u-axis. Furthermore, since the moduli of the points on the edge containing 0 and $1 + i$ vary from 0 to $\sqrt{2}$, the moduli of the images of these points vary from $0^2 = 0$ to $\left(\sqrt{2}\right)^2 = 2$. Thus, the image of this side is a vertical line segment from 0 to $2i$ contained in the v-axis and shown in black in **FIGURE 2.4.6(b)**. In a similar manner, we find that the image of the side of S containing the vertices 0 and $1 - i$ is a vertical line segment from 0 to $-2i$ contained in the v-axis. See Figure 2.4.6. The remaining side of S contains the vertices $1 - i$ and $1 + i$. This side consists of the set of points $z = 1 + iy$, $-1 \leq y \leq 1$. Because this side is contained in the vertical line $x = 1$, it follows from (2) and (3) of Example 2 that its image is a parabolic segment given by:

$$u = 1 - \frac{v^2}{4}, \quad -2 \leq v \leq 2.$$

Thus, we have shown that the image of triangle S shown in color in Figure 2.4.6(a) is the figure S' shown in black in Figure 2.4.6(b). ☰

The Function z^n, $n > 2$ An analysis similar to that used for the mapping $w = z^2$ can be applied to the mapping $w = z^n$, $n > 2$. By replacing the symbol z with $re^{i\theta}$ we obtain:

$$w = z^n = r^n e^{in\theta}. \tag{5}$$